

Aishwarya Hareesh Rao

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Embedded Systems · Firmware & RTOS · TinyML · Edge AI · VLSI Design

EDUCATION

Arizona State University
M.S. Electrical Engineering

Tempe, Arizona
Jan 2025 – Dec 2026

Visvesvaraya Technological University
B.E. Electronics & Communication Engineering

Belagavi, India
Aug 2020 – Aug 2024

SKILLS

Languages: Python, C, C++, Embedded C, MATLAB, Verilog, VHDL, System Verilog

AI / ML & TinyML: TensorFlow, TFLite Micro, PyTorch, scikit-learn, Model Quantization, CNNs, Transfer Learning

Computer Vision & Signal Processing: OpenCV, dlib, Facial Landmark Detection, Real-Time Video Processing, Signal Processing, Time-Series Analysis

Embedded & Robotics: Arduino, ARM Cortex-M4, ROS2, Gazebo, rviz, Sensor Fusion, LiDAR, RGB-D Camera, Point Cloud Processing, PCB Design & Fabrication

VLSI & EDA: Cadence Virtuoso, HSPICE, StarRC, DRC/LVS Verification, Full-Custom Layout, Post-Layout Timing Analysis, Parasitic Extraction, Heterogeneous Integration, Semiconductor Packaging

Tools: Git, MS Excel, MS Word and Powerpoint, Linux, Jupyter, VS Code, ANSYS

EXPERIENCE

Center for Artificial Intelligence and Robotics (DRDO)

Bangalore, India

Project Intern

Sep 2023 – Nov 2023

- Developed a multi-sensor fusion pipeline for autonomous ground vehicle object detection by integrating **LiDAR (360° FoV, 0.2–100 m range)**, **RGB-D camera (0.1–30 m depth)**, and Radar data streams using ROS2 and **rviz** for real-time 3D point cloud visualization.
- Built and validated custom **UGV simulation** environments in **Gazebo** with 3D robot models designed in **Blender**, enabling repeatable autonomous navigation testing across cluttered obstacle scenarios without physical hardware dependency.
- Implemented **ROS2** sensor data synchronization across 3 modalities (LiDAR, camera, radar) using **tf2 transforms** and topic-level message filtering, enabling time-aligned fusion for reliable obstacle detection and environmental mapping at **30 FPS** camera input.

PROJECTS

STM32 Secure Bootloader & OTA Firmware Engine

Feb 2026 · Independent

- Developed a fault-tolerant STM32 bootloader in bare-metal C implementing a 6-state boot decision state machine, achieving 100% firmware integrity verification via **CRC32** across a **512KB** flash layout with safe fallback to update mode on corruption, validated across all three boot scenarios (blank flash, successful update, CRC mismatch).
- Implemented a **UART** firmware update protocol from scratch, transferring firmware images in **128-byte** packets with per-packet **CRC16** sequence validation and **ACK/NACK** retransmit logic, achieving 100% packet delivery, with a Python host updater tool and full staging-to-app flash copy pipeline targeting **ARM Cortex-M4** vector table relocation.

IMU-Based Posture Detection System

Aug - Dec 2025 · ASU

- Achieved **99.99% classification accuracy** across 5 posture classes by training feedforward and **1D-CNN** models on **9-axis** IMU data with z-score normalization, Gaussian augmentation, and sliding-window feature extraction (100 samples, 50% overlap).
- Deployed **3 sensor-specific models** (~6K parameters, 19-65 KB) to an ARM Cortex-M4 via TFLite Micro, delivering **sub-150ms** on-device inference with no cloud dependency within a **32 KB** arena through an automated Keras-to-C-header conversion pipeline.

Integrate-and-Fire Neuron

Jan - May 2025 · ASU

- Achieved **391 μm^2 layout area**, **5.79 ps worst-case propagation delay**, and **0.5 ns minimum clock period** by designing a digital integrate-and-fire neuron from gate-level schematic through full-custom physical layout in **32 nm CMOS** using **Cadence Virtuoso**, validated through HSPICE transient simulation and StarRC parasitic extraction across pre- and post-layout netlists.
- Delivered **DRC-clean, LVS-passing sign-off** for both the IF neuron and D flip-flop by applying transistor-pair matching, signal shielding, and parasitic-aware routing in Cadence Virtuoso, eliminating all design-rule violations and achieving timing closure under worst-case parasitic loading.

Driver Distraction & Drowsiness Detection System

Jan - Aug 2024 · RVITM

- Achieved **86% detection accuracy** across **1,600 observations** by building a real-time computer vision pipeline that extracted Eye Aspect Ratio (EAR), PERCLOS, gaze ratio (pupil-to-corner displacement), and 3-axis head pose estimation (yaw, pitch, roll) from **30 FPS** dashcam video using facial landmark detection.
- Reduced driver alert response time to **under 3 seconds** by building a closed-loop control system that fused drowsiness signals (**EAR < 0.20 for $\geq 2.5\text{s}$**) with gaze deviation tracking to trigger buzzer actuation and automated SMS notifications, with no cloud dependency.